

Title of the Manuscript

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10 ABSTRACT: Enter the text of your abstract here. This is a sample American Meteorological
11 Society (AMS) L^AT_EX template. This document provides authors with instructions on the use of
12 the AMS L^AT_EX template. Authors should refer to the file amspaperV6.2.tex to review the actual
13 L^AT_EX code used to create this document. The templateV6.2.tex file should be modified by authors
14 for their own manuscript.

15 SIGNIFICANCE STATEMENT: Enter significance statement here, no more than 120
16 words. See [www.ametsoc.org/index.cfm/ams/publications/author-information/
17 significance-statements/](http://www.ametsoc.org/index.cfm/ams/publications/author-information/significance-statements/) for details.

18 **1. Introduction**

19 This document will provide authors with the basic American Meteorological Society (AMS)
20 formatting guidelines. This document was created using L^AT_EX and demonstrates how to use the
21 L^AT_EX template when submitting a manuscript to the AMS. The following sections will outline the
22 guidelines and formatting for text, math, figures, and tables while using L^AT_EX for a submission
23 to the AMS. An attempt to compile `amspaperv6.2.tex` should be made before using the template.
24 The files have been tested using T_EX Live (available online at <http://www.tug.org/texlive/>).
25 Feedback and questions should be sent to latex@ametsoc.org. Additional information is available
26 on the AMS L^AT_EX Submission Info web page (www.ametsoc.org/pubslatex).

27 Authors should use the empty `templatev6.2.tex` to begin their paper. A valuable source of L^AT_EX
28 information is the TeX Frequently Asked Questions page (available online at [https://texfaq.
29 org/](https://texfaq.org/)).

30 **2. Formatting text and sections**

31 The text should be divided into sections, each with a separate heading and consecutive numbering.
32 Note, however, that single secondary, tertiary, and quaternary sections remain unnumbered. Each
33 section heading should be placed on a separate line using the appropriate L^AT_EX commands.

34 *Secondary headings*

35 Secondary headings labeled with letters are formatted using the `\subsection*{}` for a single
36 subsection within a section or `\subsection{}` for multiple subsections within one section.

37 TERTIARY HEADINGS

38 Tertiary headings are formatted using the `\subsubsection*{}` for single a subsubsection within a
39 subsection or `\subsubsection{}` for multiple subsubsections within a subsection.

40 *Quaternary headings* Quaternary headings are formatted using the `\paragraph*{}` for a single
41 paragraph within a subsubsection or `\paragraph{}` for multiple paragraphs within a subsection.

42 **3. Citations**

43 Citations to standard references in text should consist of the name of the author and the year
44 of publication, for example, Becker and Schmitz (2003) or (Becker and Schmitz 2003) using the
45 appropriate `\citet` or `\citep` commands, respectively. A variety of citation formats can be used
46 with the `natbib` package; however, the AMS prefers that authors use only the `\citet` and `\citep`
47 commands. References should be entered in the `references.bib` file. For a thorough discussion of
48 how to enter references into the `references.bib` database file following AMS style, please refer to
49 the **AMS_RefsV6** document included in this package.

50 **4. Formatting math**

51 The following sections will outline the basic formatting rules for mathematical symbols and
52 units. In addition, a review of the `amspaper.tex` file will show how this is done with the use of
53 $\LaTeX$ commands. The AMS template provides the American Mathematical Society `math`, `font`,
54 `symbol`, and `boldface` packages for use in math mode.

55 *a. Mathematical symbols*

56 Symbols must be of the same font style both in text discussion and in displayed equations or
57 terms (and figures should be prepared to match). Scalar single-character symbols are set italic,
58 Greek, or script. Examples are u , L [note that ν (Greek upsilon) is used instead of v (italic “vee”)
59 to avoid confusion with ν (Greek nu) often used for viscosity; this is handled automatically when
60 in $\LaTeX$ math mode], w , x , y , z , f , g , r , indices such as i or j , and constants such as C_D , k , or
61 K . Multiple-character scalar variables, abbreviations, nondimensional numbers, and acronyms for
62 variables are set regular nonitalic: LWC, Re, Ro, BT, abs, obs, max, min, Re/Im (real/imaginary),
63 etc. For vectors, use boldface nonitalic Times Roman as in $\mathbf{V}$, $\mathbf{v}$, or $\mathbf{x}$, and $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ unit vectors.
64 Do not use the $\LaTeX$ `\vec` command to denote vectors. For matrix notation, use nonitalic boldface
65 Arial (or sans serif) font as in $\mathbf{A}$, $\mathbf{B}$, or $\mathbf{M}$. All mathematical operator abbreviations/acronyms

are set lowercase regular Roman font, except O (on the order of): $\sin$, $\cos$, $\tan$, $\tanh$, cov , Pr (for probability; note same as Prandtl number), const (for constant), c.c. (complex conjugate).

b. Units

Units are always set on a single line with a space separating the denominator, which is set with a superscript -1 , -2 , and so on, rather than using a slash for “per.” Examples are g kg^{-1} , $\text{m}^2 \text{s}^{-1}$, W m^{-2} , g m^{-3} , and m s^{-1} (note that ms^{-1} is the unit for “per millisecond”).

c. Equations

Brief equations or terms set inline in text must be set as a single-line expression because page proofs are not double spaced, for example, $\rho^{-1}p/x$ or $(1/\rho)p/x$ or $(a-b)/(c+d)$; that is, use a superscript -1 for the denominator. In case of a more complicated term or equation, it should be set as an unnumbered display equation, such as

$$x = \frac{2b \pm \sqrt{b^2 - 4ac}}{2c}.$$

Otherwise, numbered display equations can be entered using the appropriate equation command, such as

$$x = \frac{2b \pm \sqrt{b^2 - 4ac}}{2c}. \tag{1}$$

Lists of equations are punctuated as written English, and commas, semicolons, and periods are placed where appropriate. Conjunctions such as “and,” “while,” “when,” or “for” are also typically placed before the final element in a mathematical phrase, as befits the intended mathematical meaning.

5. Figures and tables

a. Figures

The insertion of a sample figure (Fig. 1) and caption is given below (in the .tex document). Standard figure sizes are 19 (one column), 27, 33, and 39 (two columns) picas.

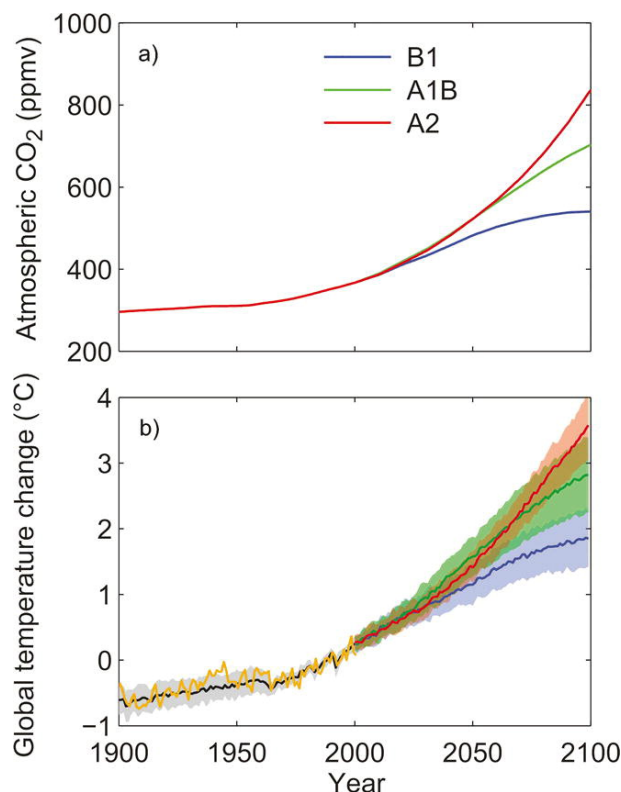


FIG. 1. Enter the caption for your figure here. Repeat as necessary for each of your figures. Figure from Knutti et al. (2008).

b. Tables

Each table must be numbered, provided with a caption, and mentioned specifically in the text. See below for sample table formatting (Tables 1 and A1). Tables cannot have underlining, color, shading, or any graphics embedded.

TABLE 1. This is a sample table caption and table layout.

<i>N</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
0000	0000	0010	0000
0005	0004	0012	0000
0010	0009	0020	0000
0014	0010	0029	0005

Acknowledgments. Keep acknowledgments (note correct spelling: no “e” between the “g” and “m”) as brief as possible. In general, acknowledge only direct help in writing or research. Financial

support (e.g., grant numbers) for the work done, for an author, or for the laboratory where the work was performed must be acknowledged here rather than as footnotes to the title or to an author's name. Contribution numbers (if the work has been published by the author's institution or organization) should be placed in the acknowledgments rather than as footnotes to the title or to an author's name.

Availability statement. The Availability Statement is where authors should describe how the data and software underlying the findings within the article can be accessed and reused. Authors should attempt to provide unrestricted access to all data and materials underlying reported findings. If data access is restricted, authors must mention this in the statement. See <http://www.ametsoc.org/PubsDataPolicy> for more info.

APPENDIX A

Title of Appendix

Appendix section

The AMS template allows authors to format an unlimited number of appendixes. [Note: AMS follows the Chicago Manual of Style, which uses "appendixes" as the plural instead of "appendices."] To format a single appendix, use the `\appendix` command with no additional argument. Otherwise, add the appropriate one-letter argument to the `\appendix` command (e.g. `\appendix[A]`, `\appendix[B]`, `\appendix[C]`, etc.) corresponding to the appropriate appendix.

The title of the appendix can be formatted using the `\appendixtitle{}` command. The `\subsection`, `\subsubsection`, and `\paragraph` commands are used to create sections within the appendix. (Note that the appendix title takes the place of `\section` in the appendix, so the first section should begin with `\subsection` instead of `\section`.) Equations are automatically numbered appropriately for each appendix. Here is an example of the first equation in appendix A, automatically labeled (A1):

$$x = \frac{2b \pm \sqrt{b^2 - 4ac}}{2c}. \quad (\text{A1})$$

Appendix figures and tables are now numbered automatically using the standard commands [i.e., the special `\appendcaption` command that was necessary in v5 has been omitted.] (Figs. A1 and A2 and Table A1).

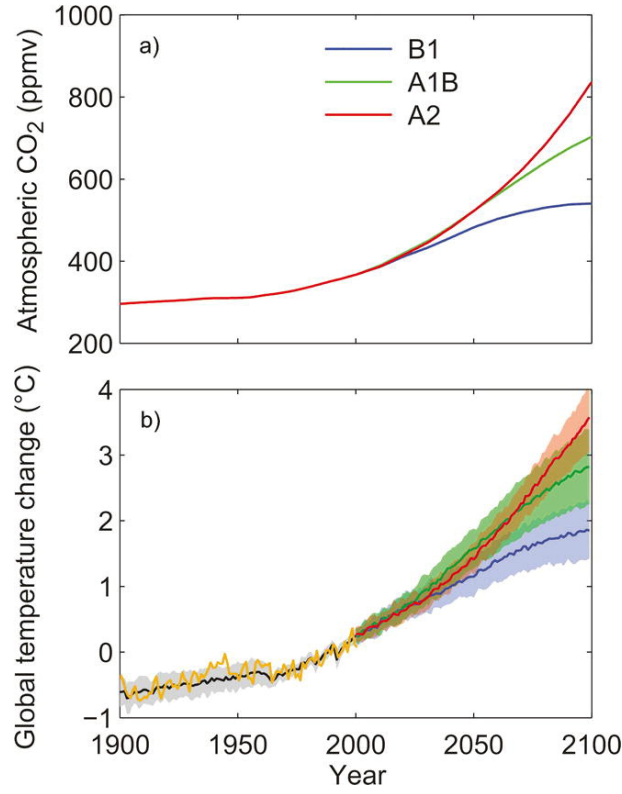


FIG. A1. Here is the figure caption for Fig. A1.

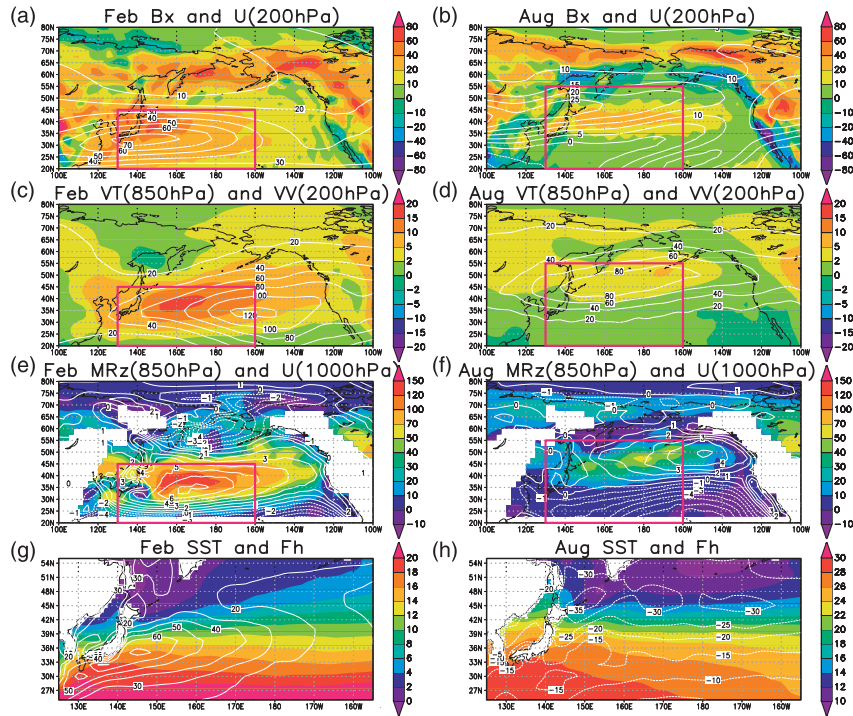


FIG. A2. Here is the figure caption for Fig. A2.

<i>N</i>	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
25	1	10	0.60	100	4	80	0.02
50	2	20	1.70	100	4	80	0.02
75	3	40	2.44	100	4	80	0.02
100	4	80	0.02	100	4	80	0.02
100	4	80	0.02	100	4	80	0.02
100	4	80	0.02	100	4	80	0.02
100	4	80	0.02	100	4	80	0.02

TABLE A1. This is sample Table A1.

Appendix tables and figures should appear in the appendix in which they are placed and not float into other appendices or the references. Place a `\clearpage` command at the end of each appendix if you note appendix tables or figures floating into other appendices.

APPENDIX B

File Structure of the AMS $\LaTeX$ Package

a. AMS $\LaTeX$ files

You will be provided with a tarred, zipped $\LaTeX$ package containing 17 files. These files are

Basic style file: `ametsocV6.2.cls`.

The file `ametsocv6.2.cls` is the manuscript style file.

- Using `\documentclass{ametsocv6.2}` for your `.tex` document will generate a PDF that follows all AMS guidelines for submission and peer review.
- Using `\documentclass[twocol]{ametsocv6.2}` for your `.tex` document can be used to generate a PDF that closely follows the layout of an AMS journal page, including single spacing and two columns. This journal style PDF is only for the author's personal use, and any papers submitted in this style will not be accepted.

Always use `\documentclass{ametsocv6.2}` when generating a PDF for submission to the AMS.

Template: `templatev6.2.tex`, for the author to use when making their paper. The file provides a basic blank template with some section headings for authors to easily enter their manuscript.

Sample .tex and .pdf files: The file `amspaperv6.2.tex` contains the $\text{\LaTeX}$ code for the sample file. The resulting PDF can be seen in `amspaperv6.2.pdf` (this file).

Sample article: Article formatted in draft and two-column mode.

- `amssamp1v6.2.tex`, `amssamp1v6.2.pdf`

Formal paper done in draft mode and the resulting .pdf.

- `amssamp2v6.2.tex`, `amssamp2v6.2.pdf`

The same paper using the `[twocol]` option and the resulting .pdf.

- `FigOne.pdf`, `FigTwo.pdf`, and `figure01.pdf` are sample figures.

Bibliography Files: `ametsocV6.bst`, `database2020.bib`, and `references.bib`.

- `ametsocV6.bst` is the bibliography style file.
- `database2020.bib` is an example of a bibliographic database file.
- `references.bib` should be altered with your own bibliography information.

Documentation: found in `AMSDocsV6.2.pdf`. Additional information found in `readme.txt`, which contains a list of the files and how they are used.

b. Help for Authors

Questions and feedback concerning the use of the AMS $\text{\LaTeX}$ files should be directed to `latex@ametsoc.org`. Additional information is available on the AMS $\text{\LaTeX}$ Submission Info web page (www.ametsoc.org/pubslatex).

APPENDIX C

Building a PDF and Submitting Your $\text{\LaTeX}$ Manuscript Files to the AMS

a. Building your own PDF

There are a variety of different methods and programs that will create a final PDF from your $\text{\LaTeX}$ files. The easiest method is to download one of the freely available text editors/compilers such as `TexWorks` or `TeXnicCenter`. `TexWorks` is installed with the $\text{\TeX}$ Live distribution and provides both a text editor and the ability to compile your files into a PDF.

166 *b. Submitting your files to the AMS for peer review*

167 The AMS uses the Editorial Manager system for all author submissions for peer review. Editorial
168 Manager uses the freely available T_EX Live distribution. This system will automatically generate
169 a PDF from your submitted L^AT_EX files and figures.

170 You should not upload your own PDF into the system. If the system does not build the PDF from
171 your files correctly, refer to the AMS L^AT_EX FAQ page first for possible solutions. If your PDF still
172 does not build correctly after trying the solutions on the FAQ page, email latex@ametsoc.org for
173 help.

174 *c. Other software*

175 As mentioned above, there is a variety of software that can be used to edit .tex files and build a
176 PDF. The AMS does not support L^AT_EX-related WYSIWYG software, such as Scientific Workplace,
177 or WYSIWYM software, such as LyX. T_EX Live (available online at
178 <http://www.tug.org/texlive/>) is recommended for users needing an up-to-date L^AT_EX distri-
179 bution with software that includes an editor and the ability to automatically generate a PDF.

180 This shows how to enter the commands for making a bibliography using BibTeX. It uses refer-
181 ences.bib and the ametsocV6.bst file for the style.

182 **References**

183 Becker, E., and G. Schmitz, 2003: Climatological effects of orography and land–sea heating
184 contrasts on the gravity wave–driven circulation of the mesosphere. *J. Atmos. Sci.*, **60**, 103–118,
185 [https://doi.org/10.1175/1520-0469\(2003\)060<0103:CEOOAL>2.0.CO;2](https://doi.org/10.1175/1520-0469(2003)060<0103:CEOOAL>2.0.CO;2).

186 Knutti, R., and Coauthors, 2008: A review of uncertainties in global temperature projections over
187 the twenty-first century. *J. Climate*, **21**, 2651–2663, <https://doi.org/10.1175/2007JCLI2119.1>.